

Data Capture and ERP Integration for the Battery Disassembly Line

Deep discharge to powder stripping: measurement points, recording conventions and system interfaces

Line: module deep discharge → manual disassembly → casing removal → electrode separation → powder stripping
Status: procurement-phase revision. To be updated once equipment interface parameters are confirmed.

1. Purpose and scope

This document covers the data capture design and the ERP interface design for a lithium battery module disassembly line. The line is currently in the equipment procurement phase and software development has not begun.

Two categories of decision become substantially more expensive to implement after equipment delivery: **the openness of the equipment data interfaces** and **the weighing equipment installed at each station**. The first is determined by purchase contract terms, the second by line layout design. Neither falls within software development. The document is therefore issued during procurement and serves three purposes: technical clauses for procurement negotiation, the measurement point basis for instrumentation and station layout, and the foundation for ERP-side interface design once equipment is on site.

No specific threshold values are given. Alarm temperatures, discharge end voltages and permitted deviation bands must be derived from equipment vendor documentation, local regulation and site work instructions. A general specification has no basis for setting them. What this document defines is the set of measurements, the reason each is required, and where the resulting data belongs.

2. Data layering principle

A disassembly line produces three categories of data with materially different characteristics. Sampling rate, purpose and storage requirements differ across all three, and they are handled in separate layers rather than combined into one channel.

Category	Typical rate	Purpose	Layer
Safety and interlock Temperature, flammable and toxic gas, smoke, emergency stop state, insulation monitoring	Milliseconds to seconds	Halt equipment, start extraction and trigger evacuation before an incident	Site layer. Hardwired to alarm, extraction and interlock circuits. The ERP receives exception events only, never the continuous stream
Mass balance Weighing, counting and sorting decisions at each station	Once per batch or once per station	Cost accounting, yield statistics, hazardous waste traceability	ERP layer. Constitutes the core business data of the ERP
Quality and assay Powder metal content, moisture, particle size, foil purity	Sampled per batch	Pricing, downstream process parameters, external certification	ERP layer assay records. The document type already exists in the system

2.1 Rationale for excluding millisecond data from the ERP database

A single sensor sampling at 100 Hz produces approximately 8.64 million rows per day; ten sensors produce approximately 86 million rows per day. The entire existing ERP holds a three-figure number of business document rows. The two differ by more than six orders of magnitude.

Writing high-frequency data into the same database would cause backups, schema verification, mirror comparison and existing check scripts to fail within a short period, and would degrade the performance of unrelated business queries. The business system and the process data system would then be constrained by a single bottleneck, with capacity problems in either affecting the other.

The layering is as follows. The site layer performs capture, buffering and aggregation. The ERP layer receives only aggregated, conclusive data. Raw waveforms remain in the site layer, with the ERP holding a locating pointer consisting of equipment identifier and time interval. Retrieval of a specific moment is performed against the site layer using that pointer.

The ERP is not a running dependency of the production line. During network outage or ERP maintenance, the line continues to operate, data queues in the site layer and is transmitted in sequence once connectivity returns. A design that makes line operation dependent on ERP availability will be bypassed by offline methods at the first network fault, at which point the data chain is broken.

3. Measurement points and recording requirements by stage

3.1 Deep discharge (Bosch equipment)

Deep discharge is the only stage on the line whose readings constitute an admission condition for subsequent stages. Casing removal, electrode separation and powder stripping all proceed on the premise that incoming cells have been discharged. Where that premise fails, the consequence is fire or thermal runaway, which cannot be remedied by rework.

Measurements

- **Residual voltage at the outlet, captured per module.** Batch averages and sampling are not used. An average cannot expose a single incompletely discharged module, and the risk is carried by precisely that individual unit.
- **Discharge duration and the terminal form of the discharge curve.** A normal module reaches its end voltage within the expected window. Significantly extended duration or voltage rebound indicates an internal abnormality.
- **Module surface temperature and cabinet ambient temperature.** The criterion is **rate of temperature rise** rather than absolute temperature. The precursor signature of thermal runaway is accelerating rise, not an elevated reading.
- **Energy recovered**, for regenerative units. This is included in cost accounting and also serves as a cross-check on discharge completeness.
- **The mapping between discharge channel number and module number.** Without it, the per-module data above cannot be attributed to a physical unit.

Measurement point layout

- Voltage is measured by the equipment at the discharge channel terminals. No additional sensor is required. The implementation issue for this item is data export permission, addressed in section 8.
- Temperature: contact thermocouple or infrared measurement on the module surface, with an ambient probe inside the cabinet. Where the equipment already captures module temperature, this is again a matter of export permission.
- **Gas detection for electrolyte vapour and hydrogen fluoride.** The two differ in density and dispersion behaviour. Detector mounting height must be determined by gas properties and building ventilation arrangement, specified by a qualified safety engineering firm.
- Gas detectors are **connected directly to alarm and extraction circuits, with no software system in the path.** The ERP receives the record that an alarm occurred and carries no responsibility for triggering it.

Records entering the ERP

- Per module: module identifier, outlet voltage, admission verdict, verdict timestamp.
- Per batch: module count and mass in, total discharge duration, energy recovered, count of failed modules and their disposition.
- Exceptions: recorded individually, with time of occurrence, type, duration, action taken and responsible person.

Disposition of failed modules is a mandatory field. A module that has not completed discharge enters one of two paths, re-discharge or quarantine, and both paths generate a record. Without this field, failed modules within a batch have no traceable destination in subsequent records.

3.2 Manual disassembly (module to cell)

This stage has no process sensors. Data originates from weighing equipment and operator records. Data completeness depends on operators performing weighing before and after each job, making it the stage most exposed to data discipline failure.

Records

Record	Capture method	Necessity
Module mass and count in	Full batch weighed and counted at loading	The reference quantity for this stage's mass balance
Cell mass and count out	Weighed and counted at unloading	Downstream routes by unit ; mass alone cannot determine routing
Structural part and housing mass out	Weighed after sorted collection	The largest term in the balance; the stage cannot close without it
Harness, BMS board and busbar mass	Collected separately and weighed	High copper content; a revenue item rather than waste
Cells damaged during disassembly	Counted by the operator at the station	Reflects tooling and technique; also a safety indicator
Operator, job start and end time	Recorded automatically by the station terminal	Traceability requirement; also the source data for labour and throughput

Measures securing weighing compliance

Data completeness at this stage is secured through process design rather than administrative instruction:

1. **Weighing and recording occur within the same action.** The station is equipped with a tablet or panel PC and the entry is made at the point of weighing. Paper records with subsequent batch entry are not used; the transcription step from paper to system is the principal source of reconciliation discrepancies.
2. **Weighing equipment has a communication interface and readings populate automatically.** Operators perform confirmation and batch code scanning only. Manual key entry produces order-of-magnitude errors, such as 12.5 recorded as 125, which typically surface at monthly consolidation when the source can no longer be traced.
3. **Commencement of the next stage is conditional on completion of this stage's record.** Where weighing has not been performed, the next batch code cannot be generated.

3.3 Casing removal (separate equipment for hard case and pouch)

This stage is preceded by a classification action: determining whether each cell is of hard case or pouch construction. That determination is itself a record item and is not treated as inherent routing.

Recording requirements for the classification action

- Where classification is visual and manual: record the hard case and pouch counts determined for the batch, and the person who determined them.
- Where the equipment performs identification: record the results and **the count of units the equipment could not identify and which required manual adjudication**. The latter is the indicator of identification effectiveness.
- Under either method, **record occurrences of misclassification**. A pouch cell entering hard case equipment is normally evident immediately. A persistently non-zero figure indicates the classification step requires revision.

Process records

Record	Measurement location	Indicates
Cell mass and count in, by hard case and pouch	Loading station scale	Reference quantity for the stage
Casing mass out	Floor scale or hopper scale beneath the casing collection point	Aluminium and steel recovery, a direct revenue item
Opened cell mass and count out	Unloading station scale	Actual input quantity for the next stage
Scrap count: cut through, internal short circuit, smoke, fire	Counted by the operator, corroborated by station smoke detection records	The primary indicator of cutting parameter suitability. A rising scrap rate typically precedes visible tool failure
Cutting station temperature and smoke detection state	Adjacent to the blade and within the collection chamber	Safety interlock; also corroborating evidence for scrap cause
Cumulative cut count per tool and replacement records	Machine counter or manual log	Used to correlate scrap rate with tool life

Casing removal carries the highest mechanical and fire risk on the line concurrently. The blade may contact a tab that has not been fully discharged, and residual electrolyte within the casing presents a combustion risk in the presence of a spark. Smoke detection and suppression provisions for this stage fall within safety engineering, to be designed by a qualified firm. Selection is outside the scope of this document.

3.4 Electrode separation (separate equipment for wound and stacked)

This stage is a three-way split. A single opened cell yields cathode sheet, anode sheet and separator, together with recovered or evaporated electrolyte. The casing is removed at the preceding stage and is not an output here. Mass balance closure is most difficult at this stage.

Required material attribute addition

Routing to one of two machines is determined by **wound or stacked construction**. The existing ERP material dictionary contains thirteen physical forms — whole pack, module, loose cells, black mass, electrode scrap and others — but **does not include a wound or stacked construction dimension**. Without it the system cannot record the equipment routing for a batch, and routing depends on verbal communication on the floor.

The attribute belongs to the **cell batch** rather than the material master. Batches of the same material category, such as loose cells, may differ in internal construction depending on source. It is a property of the batch, not of the category.

Process records

Record	Purpose	Note
Cell mass, count and construction type in	Reference quantity for the stage	Construction type determines equipment routing
Cathode sheet mass out	Input quantity for the next stage	One of three outputs

Anode sheet mass out	Input quantity for the next stage	One of three outputs
Separator mass out	Plastics recovery revenue	One of three outputs
Cross-contamination rate: proportion of anode sheet within the cathode stream and the reverse	Directly affects black mass sale price. Contaminated material is subject to downgrading or additional purification cost	Sampled weighing or manual separation and weighing, per shift or per batch
Electrolyte recovered or evaporation loss estimate	A mass balance term; also environmental reporting data	Measured where recovery equipment is installed; otherwise derived from mass loss and marked as a derived value
Run time, energy consumption, unplanned stop count and duration	Cost accounting and equipment effectiveness analysis	Read from the machine controller

Cross-contamination is the only quality indicator at this stage that does not affect the mass balance. When it deteriorates the balance still closes normally, and the problem appears at the customer's assay in the form of a price deduction. Sampling frequency is not less than once per shift, with results attached to the batch so that separation quality can be traced after the material is sold.

3.5 Powder stripping (cathode and anode processed separately)

This stage forms the saleable product and carries the highest measurement precision requirement on the line. Preceding stages yield intermediates; the black mass and copper and aluminium foil produced here are sold directly.

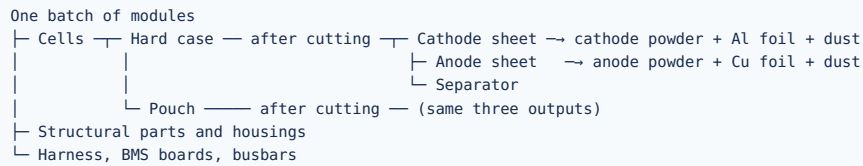
Indicator	Definition	Purpose
Powder yield	Powder mass produced per unit mass of electrode sheet	Baseline recovery indicator
Residual powder on foil	Powder remaining on the foil surface after stripping, as a proportion of theoretical powder mass	A direct loss item. Also the primary indicator of stripping parameter suitability
Foil purity	Powder and other contamination within the copper and aluminium foil	Contaminated foil cannot be priced as pure metal. Mutually constrained with residual powder: excessive stripping intensity damages the foil, insufficient intensity leaves powder
Moisture content	Water content of the powder	Affects storage safety and downstream metallurgical processing; typically a customer acceptance criterion
Particle size distribution	Size distribution of powder particles	A key parameter for downstream hydrometallurgy. Recorded continuously where in-line measurement is fitted, otherwise sampled per batch
Dust collection mass	Powder collected by baghouse or cyclone	Included in the mass balance. Where excluded, this quantity falls into unexplained loss despite being recoverable product
Energy consumption and run time	Energy per unit output	Cost accounting; also an indirect indicator of equipment condition

Measurement point layout

- **Three scales on the output side:** powder collection hopper, foil collection point and dust collection outlet. All three are required; the stage balance cannot close if any is omitted.
- **Residual powder and foil purity are obtained by sampling and assay**, not by in-line measurement. Sampling point and frequency are fixed in the work instruction, with sample identifiers bound to the batch.
- Where an in-line moisture meter is fitted, continuous readings remain in the site layer and a per-batch average is transmitted to the ERP. Otherwise material is sampled per batch and processed through the assay record.
- Dust concentration monitoring falls within safety scope owing to combustible dust risk, is hardwired to interlock circuits, and does not pass through the ERP.

4. Batch tree and mass balance

The records from the five stages form a branching tree. A single batch of modules processed through the full line yields eight to nine material categories, each with its own batch number, destination and price. The batch tree is the actual structure of this data set.



4.1 Closure principle

At each branch point, input mass equals the sum of all output masses plus named losses. This is the core constraint of the specification, and the remaining provisions exist to support it.

- **Both input and output are weighed at every branch point,** for every batch, without sampling. Where any single weighing is omitted, the balance for that stage cannot be calculated.
- **Loss items are named, with values drawn from a fixed list rather than free text.** Dust loss, electrolyte evaporation, sweepings recovered, sampling consumption and material retained in equipment are all auditable categories. A loss designated as “other”, or existing only as a difference figure, has no audit value.
- **Permitted deviation bands are defined in advance and enforced by the system.** A batch outside the band cannot be closed directly and requires a recorded explanation. Bands are set per stage rather than as a single line-wide value.

4.2 Compliance and commercial significance

Lithium battery recycling is a hazardous waste processing activity. The primary question in a regulatory inspection concerns the disposition of hazardous waste after receipt. A batch tree that closes constitutes a complete answer. A tree that does not close requires an improvised explanation of the discrepancy during inspection, which carries the lowest evidentiary weight.

The same applies commercially. Black mass purchasers generally require a statement of feedstock composition and separation quality. Tracing back along the batch tree to the supplier records and discharge records of specific module batches constitutes verifiable commercial evidence.

Process records are append-only. The reasoning matches that applied to the approval log: a process record that can be modified after the fact has no audit value. Where correction is required, a new correcting record is created referencing the original, and the original content is not modified.

5. ERP record structure

Each stage corresponds to one **processing record** in the ERP. The document type exists in the system and comprises inputs, outputs and costs. This line adds the field groups below.

Field group	Content	Source
Inputs	Source batch numbers, one or more; mass and count per batch; material type; construction type	Previous stage output; station scanning
Outputs	New batch number, mass, count and material type per output stream	Station scales; batch numbers generated by the system
Losses	Mass, reason selected from a fixed list, remarks	Mass calculated from the balance; reason selected by the operator
Process indicators	Start and end time, equipment identifier, run time, average and peak temperature, energy consumption, scrap count	Aggregated per batch by the site gateway and transmitted
Exception events	Recorded individually: time, type, duration, action taken, responsible person	Equipment alarms; manual entry
Quality	Sample identifier, assay record number, key indicator results	Linked to existing assay records
Raw data pointer	Equipment identifier, time interval, site system dataset reference	Generated by the gateway. The ERP does not store raw waveforms
Personnel and time	Operator, confirming person, record creation time, confirmation time	System-generated; station login identity

5.1 Batch numbering

The existing document numbering scheme uses a prefix-year-sequence format and covers more than thirty prefixes. Material batches added by this line use the same mechanism; a parallel numbering scheme is not established. Prefixes for cathode powder, anode powder, copper foil, aluminium foil, separator and the remaining outputs are determined and registered in a single pass during line construction, so that new batch types fall within existing validation coverage.

6. Data integration method

6.1 Site layer composition

An **edge gateway**, an industrial PC, is installed at the equipment and performs three functions:

1. **Capture.** Reads data from machine controllers. Most industrial equipment supports standard communication protocols and direct sensor connection is not required. Closed systems and earlier equipment models require additional instrumentation, to be confirmed during procurement.
2. **Local buffering.** Data is written to local storage first. Capture continues normally during network outage and is transmitted in sequence on restoration. The retention period for complete raw data on site is determined by storage cost and traceability requirements.
3. **Local aggregation.** High-frequency data is aggregated in the site layer into conclusive indicators: batch average temperature, peak temperature, total energy consumption, stop count and duration. **What is transmitted to the ERP is the aggregated result, not raw sample points.**

6.2 Transmission timing

- **A summary transmission at the close of each batch.** This is the principal data flow.
- **Exception events transmitted in real time.** Alarms, emergency stops and out-of-range conditions are transmitted on occurrence without waiting for batch close.
- **Heartbeat signal.** The gateway reports operating status at a fixed interval. **A missing heartbeat itself constitutes an exception requiring action** — without this mechanism, a failed gateway and a gateway with no data to report are indistinguishable within the system.

6.3 Automatic posting and operator confirmation

Data transmitted by the gateway **creates a pending draft, which becomes a formal record only after operator confirmation at the station.** Automatic direct posting is not used.

The rationale is that automatically captured data carries inherent error sources: scales not tared, batch codes scanned incorrectly, data windows lost to equipment restart, and consecutive batches processed on the same machine without system separation. **Erroneous data posted automatically has no mechanism by which it would be detected** and persists in the records until a monthly stocktake shows a discrepancy, at which point the originating batch can no longer be identified.

Operator confirmation establishes the operator as the control point for data admission. The approach matches that used in the

attendance module of the system, where an approver confirms the fact of work performed and personal accountability substitutes for technical anti-fraud measures.

The confirmation action itself is recorded: confirming person, confirmation time, and whether data was modified between arrival and confirmation. Where operators are permitted to correct a value at confirmation, such as after re-weighing, **both the original and corrected values are retained** and a reason for correction is required.

6.4 Landing table and data conversion

The gateway does not write to business tables. The data path is: the gateway writes to a **landing table**; ERP-side code performs validation and conversion; formal processing records are then generated. This arrangement serves three purposes:

- Gateway privileges can be reduced to insert rights on a single table with no read access at all.
- Changes in data format, arising from equipment or gateway version changes, affect only the conversion layer and not business table structures.
- **Data failing conversion is not lost;** it remains in the landing table with its state visible. This is a necessary provision, as a silently discarded transmission and a successful one are indistinguishable in a log.

7. Security and access control

An edge gateway on the shop floor is physically accessible: external contractors, removable storage and unlocked cabinets all constitute access paths. Its privileges are configured on the premise that the device may be compromised.

Principle	Implementation requirement
Dedicated identity	The gateway holds its own account and key, reuses no personnel account and does not use the system administration key. Each gateway holds a separate identity so that individual revocation is possible
Write only	Insert rights on the landing table only. Read access is zero — the gateway requires no business table access to perform its function
Minimum reach	No access to payroll, contracts, customer information or personal data. Compromise of a single shop floor device must not result in an enterprise data breach
Revocable	Keys must be individually revocable and replaceable without stopping the line. Where key rotation requires a production halt, the mechanism will not be exercised in practice
Encrypted in transit	The plant network is treated as untrusted. Transmission is encrypted and the server identity is verified
Logged	Each transmission records originating gateway, time and data volume. Abnormal transmission patterns, such as large transmissions outside working hours or transmissions from unregistered gateways, must be identifiable

A reverse channel is outside the scope of this specification: the ERP has no capability to command equipment. Data flow is unidirectional, from the site layer to the ERP layer. Should production instructions later need to be issued from the ERP, that capability requires an independent security assessment and is not to be added to the existing data channel.

8. Procurement-phase technical requirements

8.1 Data interface contract terms

Data interface terms are to be settled during procurement negotiation. A proportion of industrial equipment vendors keep data enclosed within their own systems. Requesting export capability after delivery generally incurs additional development charges on a delivery schedule outside the purchaser's control. Raising the requirement during negotiation carries close to zero cost.

Terms to be specified in the contract:

1. **Supported communication protocol.** A standard industrial protocol is required. Proprietary-only protocols and arrangements accessible solely through the vendor's cloud platform are not accepted.
2. **A complete list of readable parameters.** General statements such as “supports data export” are not accepted. An itemised point list is required: parameter name, unit, range, precision, update rate.
3. **Sampling rate and timestamp precision,** specified in particular for safety-related parameters.
4. **No additional charge.** The data interface is a component of the equipment, not a value-added service.
5. **Local historical data retention period and bulk export capability.**
6. **Interface documentation included in the deliverables list,** as complete documentation in English or Chinese.

One item requires separate confirmation for the deep discharge equipment: **whether per-module voltage data can be exported.** This data constitutes the safety admission basis for the entire line. Where it can be viewed only on the equipment's own interface and cannot be exported, the fact of per-module verification has no corresponding evidence within the ERP.

8.2 Weighing equipment configuration

A line without weighing provision cannot produce a mass balance, and the quantity of sensors installed does not alter that conclusion. Mass balance is the part of this data set serving cost accounting and compliance simultaneously. Scale positions, capacities and communication arrangements are determined during line layout design.

Station	Equipment type	Configuration considerations
Module loading	Floor scale or forklift scale	Capacity covers a full pallet; precision requirement is low
Manual disassembly station	Bench scale	Cells, structural parts and harness weighed separately; multiple units or sequential use of one
Casing removal, loading and unloading	Bench scale and hopper scale	Casing collection uses a hopper scale supporting continuous accumulation
Electrode separation outputs	Three bench scales	One per output stream
Powder stripping outputs	Three hopper scales	Powder, foil and dust collection, one each
Finished goods packing	Floor scale	Reweighing before dispatch, constituting the final verification for the line

Selection confirmation items:

- **A communication interface**, with readings transmitted automatically to the station terminal. Manual key entry is the principal error source.
- **Precision and capacity matched to actual batch size**. A 500 kg capacity unit weighing a 2 kg sample produces a reading of no reference value.
- **Dust and corrosion protection ratings**. A black mass environment imposes real requirements on ingress protection.
- **A scheduled calibration regime**, with calibration records held in the system. All data produced by an uncalibrated instrument is unreliable.

9. Appendix: consolidated measurement point list

All measurement points by stage. The destination column states the final storage layer.

Stage	Measurement or record	Category	Destination
Deep discharge	Outlet residual voltage, per module	Mass / quality	ERP, one row per module
Deep discharge	Discharge duration and curve terminal form	Process	Raw data site layer, summary to ERP
Deep discharge	Module surface and cabinet temperature, rate of rise	Safety	Site interlock, exceptions to ERP
Deep discharge	Electrolyte vapour and hydrogen fluoride concentration	Safety	Hardwired alarm and extraction, exceptions to ERP
Deep discharge	Energy recovered	Process / cost	ERP, per batch
Manual disassembly	Module mass and count in	Mass	ERP
Manual disassembly	Cell mass and count out	Mass	ERP
Manual disassembly	Structural part, housing, harness and BMS board mass	Mass	ERP
Manual disassembly	Cells damaged during disassembly	Quality / safety	ERP
Casing removal	Classification verdict and adjudication count	Mass	ERP
Casing removal	Cell in, casing out, opened cell out mass	Mass	ERP
Casing removal	Scrap count and cause	Quality	ERP
Casing removal	Cutting station temperature and smoke state	Safety	Site interlock, exceptions to ERP
Casing removal	Cumulative cut count per tool and replacement records	Equipment	ERP, maintenance records
Electrode separation	Cell mass, count and construction type in	Mass	ERP
Electrode separation	Cathode, anode and separator mass out	Mass	ERP, one row per stream
Electrode separation	Cross-contamination rate, sampled	Quality	ERP, attached to batch
Electrode separation	Electrolyte recovered or evaporation loss	Mass / environmental	ERP, marked measured or derived
Electrode separation	Run time, energy consumption, unplanned stops	Process / cost	ERP, aggregated per batch
Powder stripping	Powder mass out, cathode and anode separately	Mass	ERP
Powder stripping	Foil mass out, copper and aluminium separately	Mass	ERP
Powder stripping	Dust collection mass	Mass	ERP, included in the balance
Powder stripping	Residual powder on foil, sampled	Quality	ERP, assay record
Powder stripping	Foil purity, sampled	Quality	ERP, assay record
Powder stripping	Powder moisture content	Quality	ERP, assay record or per-batch average
Powder stripping	Particle size distribution	Quality	ERP, assay record
Powder stripping	Dust concentration	Safety	Hardwired interlock, exceptions to ERP
Powder stripping	Energy consumption and run time	Process / cost	ERP, per batch
Whole line	Operator, confirming person, start and end time	Traceability	ERP, every record
Whole line	Gateway heartbeat	System	ERP, absence constitutes an exception

10. Implementation sequence

1. **Procurement phase:** incorporate the technical terms of section 8 into procurement negotiation, and incorporate scale positions and models into line layout design. Both items become substantially more expensive to address later.
2. **After equipment selection:** obtain the point list for each machine and check it item by item against the appendix, establishing the distribution across three categories — directly readable, requiring additional instrumentation, and unobtainable. This step produces the first cost estimate based on actual interfaces.
3. **During delivery and commissioning:** bring scales and station terminals into use first, and **establish the mass balance on paper or in a spreadsheet before any system exists**. This phase exposes process problems while the cost of process adjustment is at its lowest.
4. **After the line reaches stable operation:** commence ERP-side development. Batch rhythm, branching structure and exception types are established by this point, and development proceeds from operating data rather than design assumption.
5. **Final phase:** connect the site gateway and automate the verified manual process.

The third step is the critical node in this sequence. The common failure path in comparable projects is to develop the system first, discover after deployment that floor practice does not match system assumptions, and then spend several months modifying the system to accommodate site reality. Verifying the process before developing the system avoids that path.